

# Yidong Zhang 张屹东

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## EDUCATION

**The Chinese University of Hong Kong**

Sep 2025 – Jul 2026 (expected)

*M.Sc. in Electronic Engineering*

**Beijing Forestry University** (“Double First-Class” / Project 211 university)

Sep 2021 – Jun 2025

*B.Eng. in Electronic and Information Technology · GPA: 86.52/100*

Relevant coursework: Advanced Mathematics, Mathematical Modeling, FPGA Practice and Applications, Embedded Systems and Applications, Machine Learning

## PUBLICATIONS

- Alaa Eldin Abdelaal, ..., **Yidong Zhang**, et al. Open-H-Embodiment: A Large-Scale Dataset for Training Foundation Models in Healthcare Robotics. *Science Robotics* (IF 32.9). Under review; co-author, authors listed alphabetically. [arXiv:2604.21017](https://arxiv.org/abs/2604.21017)
- Chikit Ng\*, **Yidong Zhang**\*, Tianchun Wu, et al. EndoVLA-OOV: Vision-Language Action Model for Out-Of-View Object Search-to-Track towards Distal Endoscopy Autonomy. *IEEE Transactions on Robot Learning* (CAS Q1). Major revision; co-first author.
- Chi Kit Ng\*, **Yidong Zhang**\*, Lui Siu Hing, et al. EndoLIFT: Language-Disambiguated Latent-Conditioned Rectified Flow for Bidirectional Endoscopic Control. *Biomimetic Intelligence and Robotics* (IF 8.6). Accepted; co-first author. [arXiv:2608.20478](https://arxiv.org/abs/2608.20478)
- Chi Kit Ng\*, **Yidong Zhang**\*, Shuchang Liu, et al. EndoSkillWeaver: Weaving Disjoint Visual-Motor Skills for Endoscopic Aperture Insertion. *CoRL 2026 Workshop*. Under review; co-first author.
- Siyu Chen, Yixuan Li, **Yidong Zhang**, Yifan Yang, and Xiangxue Zhang. Soft X-ray image recognition and classification of maize seed cracks based on image enhancement and optimized YOLOv8 model. *Computers and Electronics in Agriculture*, vol. 216, 108475, Jan 2024 (CAS Q1, IF 8.9). Third author. DOI: [10.1016/j.compag.2023.108475](https://doi.org/10.1016/j.compag.2023.108475)
- Chi Kit Ng\*, Hang Li\*, **Yidong Zhang**, et al. A Source-Level Illumination-Motion Co-Control System for Robotic Endoscopy. *ICBIR 2026*. Accepted (oral); third author.
- Yidong Zhang** and Xinghua Yang. Research on improved LSTM power transformer oil temperature prediction based on spatio-temporal fusion. In *Third International Conference on Machine Vision, Automatic Identification, and Detection (MVAID 2024)*, Proc. SPIE vol. 13230, pp. 598–604, Aug 2024. First author. DOI: [10.1117/12.3035438](https://doi.org/10.1117/12.3035438)

\* Equal contribution. CAS Q1: Tier 1 journal in the Chinese Academy of Sciences journal ranking. IF: journal impact factor.

## PATENTS

- A YOLOv8 neural-network method for recognizing internal cracks in maize seeds from soft X-ray images. Chinese invention patent application (published).
- An automatic detection device for internal cracks in maize seeds based on soft X-ray image recognition. Chinese invention patent application (published).

## RESEARCH EXPERIENCE

**NVIDIA Open-H & Intelligent Medical Mechatronics Lab**

Sep 2025 – Present

*Research Assistant*

- Developed a multi-motor endoscope teleoperation system: built a 4-axis DJI M2006 motor control system with a RoboMaster A board and a Python host program, achieved high-frequency closed-loop motor control over CAN bus, and integrated an Xbox controller for vector-based endoscope actuation and PWM lighting control.
- Designed a multimodal data acquisition pipeline (serial communication, data logging, CSV storage), resolving timestamp alignment and synchronization between variable-frame-rate video and fixed-rate sensor streams.
- Built an endoscope data-collection tool with a visual UI recording motor position, velocity and joystick control vectors; cleaned and standardized the data into imitation-learning datasets compatible with the LeRobot framework.
- Trained end-to-end imitation-learning policies with the OpenPi framework ( $\pi 0.5$  series models), configuring multi-source datasets and joint training, and improving convergence and generalization through analysis of loss curves and training logs.
- Deployed and validated an AI endoscope model on embedded hardware with real-time inference from visual and state inputs, forming a closed loop from image input to motor action output.

**Hong Kong RGC Joint-Funded Project:** Vision-Language-Navigation (VLN) Robotic System for Constrained Curved Environments

Sep 2025 – Present

- Studied autonomous navigation with multimodal large models in constrained, curved environments, targeting the complex control requirements of flexible robots.
- Built an end-to-end VLN large-model control framework; developed a dual-motor teleoperation platform and designed a multimodal data acquisition system, addressing multi-source data alignment.
- Led model development and deployment on Linux, integrating natural-language instructions, visual perception and low-level motor control to map multimodal inputs directly to motor commands.
- Outcome: precise endoscope navigation and trajectory tracking inside constrained curved channels, moving beyond the limitations of traditional geometric modeling, with potential applications in medical and other complex settings; related work is under major revision at *IEEE Transactions on Robot Learning*.

**Beijing Municipal-Level Project:** Automatic Maize Seed Damage Detection Device Based on Soft X-ray Imaging and Image Recognition

Feb 2022 – Apr 2024

- Built a soft X-ray imaging system combined with deep learning for automated, non-contact detection of internal damage in maize seeds, deployed end-to-end from visual recognition to sorting actuation.
- Responsible for system setup and model training: built and preprocessed a standardized annotated seed-damage dataset; optimized the YOLOv8 architecture and loss function to improve detection of fine cracks.
- Conducted ablation studies and metric evaluation; exported the model and integrated it with the sorting device to close the loop from recognition to actuation, supporting project acceptance and deployment.

## INDUSTRY EXPERIENCE

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**Chengshi Zhiguang (Shenzhen) Autonomous Driving Co., Ltd.** (城市之光), Perception Algorithm Dept.

Jun 2025 – Aug 2025

*Data Processing Intern*

- Annotated and cleaned cloud-based perception data for autonomous driving to ensure training-data quality.
- Assisted cloud deployment of the Sparse4D multi-camera 3D perception model via PyTorch → ONNX → TensorRT export and high-performance inference.
- Contributed to baseline training of Sparse4D/BEVFormer spatio-temporal Transformers, tuning DDP random seeds, learning-rate schedules and evaluation scripts; wrote weekly experiment reports and iteration recommendations.

**Kylin Software Co., Ltd.**, Xingguang Kylin Department

Jul 2024 – Aug 2024

*Embedded Development Intern*

- Compiled and adapted OpenWrt for the BCM2711 chip on the Linux kernel; completed functional and performance testing, flashed the system to a Raspberry Pi 4B and verified its stability.

## HONORS & AWARDS

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- Third Prize (Innovation Project), 12th Hong Kong University Student Innovation and Entrepreneurship Competition, 2026
- Second Prize (Beijing Division), National Undergraduate Electronic Design Contest, 2023
- Third Prize, 33rd Beijing College Student Mathematics Competition (Non-Mathematics Majors, Group B), 2023
- Third Prize, 8th National College Student Physics Experiment Competition; Third Prize, 15th Beijing College Student Physics Experiment Competition
- Merit Student, Beijing Forestry University (twice); University Academic Excellence Scholarship; University Outstanding Student Scholarship (Second Class and Third Class, once each)

## SKILLS

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- **ML / Robot learning:** PyTorch, imitation learning (OpenPi, LeRobot), vision-language-action models, YOLOv8, Transformer-based 3D perception (Sparse4D, BEVFormer), DDP training
- **Deployment & systems:** ONNX, TensorRT, embedded model deployment, Linux, OpenWrt
- **Hardware & embedded:** motor control over CAN bus, serial communication, PWM, RoboMaster/DJI motors, Raspberry Pi, FPGA; Python

## ACTIVITIES

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- Member, Chinese Association of Automation
- Outstanding Volunteer, National Higher-Education Knowledge Popularization Volunteer Service
- Member, Arts Department, Student Union, School of Science, Beijing Forestry University